# AlphaTrace

*An evidence-first autonomous financial research investigator*

**TRACK A**

BANKING | NBFCs | FINANCIAL SERVICES

## THE IDEA

Given only a NIFTY 50 company, AlphaTrace decides which financial relationships are worth investigating. It turns public filings, peer disclosures and independent sector evidence into one auditable discovery - not a finance chatbot, news summary or investment recommendation.

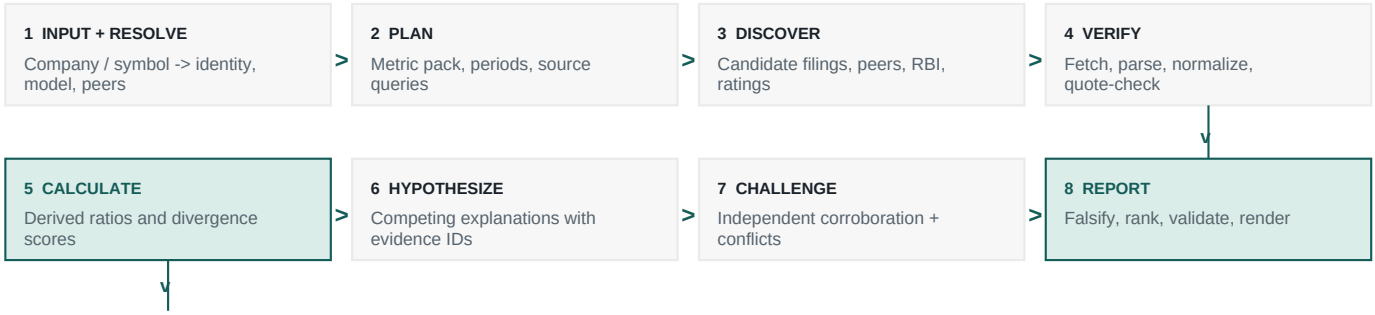
The problem is fragmentation: a company can describe improving asset quality while the flow of stress, its peers, or an external source tells a different story. A single-document summariser cannot reliably expose that relationship.

**COMPUTE-FIRST DIFFERENTIATOR**

The pipeline computes a time-indexed grid of metrics first, detects a divergence second, and asks the model to explain only validated patterns. The anomaly is found by arithmetic; prose cannot invent the number or URL.

## RESEARCH PIPELINE

The same sequential coordinator drives offline fixtures and live public-source runs. Each stage emits inspectable artefacts; rejected hypotheses and missing coverage remain visible.




*If evidence is insufficient, return to planning or report: No strong discovery found.*

## DISCOVERY MECHANISM

The first Track A pack asks whether headline asset-quality improvement is supported by the underlying flow of stress. The lens library is reusable; the resolver selects the applicable pack for Bank, NBFC, Insurer or Holdco.

- Stock vs flow: GNPA with NNPA, write-offs, credit cost or provision coverage.
- Self vs peer: distinguish a company-specific signal from sector normalization.
- Narrative vs number, segment vs aggregate, and entity vs counterparty where data permits.
- Temporal persistence: align periods, detect when the relationship changed, and test base effects.

## CONFIRMED NOW / FINALS NEXT

**CONFIRMED IN REPOSITORY**

- Offline end-to-end research engine and dashboard.
- Deterministic calculations, typed report objects, claim ledger and SQLite replay.
- Public URL validation, bounded retrieval/cache, page-aware parsing and quote verification.
- Conservative live mode that downgrades or withholds weak findings.

**FINALS WORK**

Broaden peer/source coverage, metric packs and cold unseen-company validation; keep every limitation explicit.

*Design rule: deterministic code owns arithmetic, source identity, quote status, independence metadata, conflict classification and final schema validation.*

**Public sources only | No investment recommendations | Company-specific conclusions are never hard-coded**

# Make every conclusion contestable

*Evidence triangulation, output contract, generalisation and delivery plan*

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## EVIDENCE + CONTRADICTION

Search results are candidates, not evidence. A source enters the ledger only after public-URL validation, bounded retrieval, parsing and exact or controlled-normalized quote verification.

- Every fact stores metric, value, unit, period, scope, source URL, publisher family, tier, page and exact passage.
- Every claim points to evidence IDs and calculation IDs; displayed numbers come from deterministic Python code.
- Independent-family count prevents three company-authored documents from masquerading as three confirmations.
- Typed conflicts surface narrative/metric, restatement, scope, timing and company/peer mismatches instead of hiding them.

**Evidence ladder**

| | |
|---|---|
| **T1** | NSE/BSE filings, results, annual reports |
| **T2** | Investor presentations and earnings calls |
| **T3** | RBI / IRDAI / SEBI independent data |
| **T4-T6** | Ratings, peers, counterparties, credible media |

## SELF-FALSIFICATION

**The leading hypothesis must earn its rating**

**T1** **Sector**
Do most peers show the same divergence?

**T2** **Accounting**
Could scope, M&A or definition drift explain it?

**T3** **Base effect**
Does it survive two- and three-year views?

**T4** **Macro / regulatory**
Does timing align with an industry event?

**T5** **Disclosure change**
Did the company change what it reports?

*PASS / FAIL / INCONCLUSIVE is evidence, not decoration.*

## ILLUSTRATIVE FINAL REPORT OBJECT

The renderer receives one validated object with all eight required elements. The sketch below shows the shape, not a researched company result or fabricated figure.

**1 DISCOVERY**
*[one clear non-obvious finding]*

**2 WHY IT MATTERS**
*[3-5 significance bullets]*

**3 EVIDENCE**
*[independent sources + clickable links]*

**4 TIMELINE**
*[when the change emerged]*

**5 NUMBERS**
*[formula -> inputs -> result]*

**6 COUNTERARGUMENT**
*[strongest alternative / conflict]*

**7 FALSIFICATION**
*[future condition that changes view]*

**8 STRENGTH**
*[Strong / Moderate / Weak + why]*

*A click on any major claim opens its document, page, passage, period and calculation trace.*

## GENERALITY

Constants: typed schemas, source policy, quote checks, calculation engine, conflict types, five tests and report contract. Dynamic at runtime: company identity, business model, peers, periods, source plan, metric pack and hypotheses.

A resolver classifies Bank / NBFC / Insurer / Holdco and selects a compatible pack. Missing disclosures reduce evidence strength or stop a conclusion; a new company does not require a new answer template.

## TOOLS + DATA

**STACK**

Python 3.12.8; explicit sequential coordinator; vanilla HTML/CSS/JS; SQLite; standard-library HTTP/HTML parsing; Docker Poppler boundary for PDFs; optional Tavily search and structured OpenAI-compatible provider.

*No multi-agent framework is required to make the trail auditable.*

## BUILD PLAN TO FINALS

| | |
|---|---|
| **18-19 OCT** | Freeze Round 1 report; run offline end-to-end checks; review source and limitation wording. |
| **20-21 OCT** | Broaden peer retrieval and independent-family scoring; extend bank/NBFC/insurer metric coverage. |
| **22 OCT** | Harden temporal alignment, typed contradictions and five falsification tests; preserve weak-result fallback. |
| **23-24 OCT** | Cold-run unseen NIFTY 50 and other listed Track A companies; Docker build; cache a recovery run; rehearse demo. |
| **25 OCT** | Finals: live judge-selected company, discovery, evidence trail, counterargument and falsification condition. |